

Elements

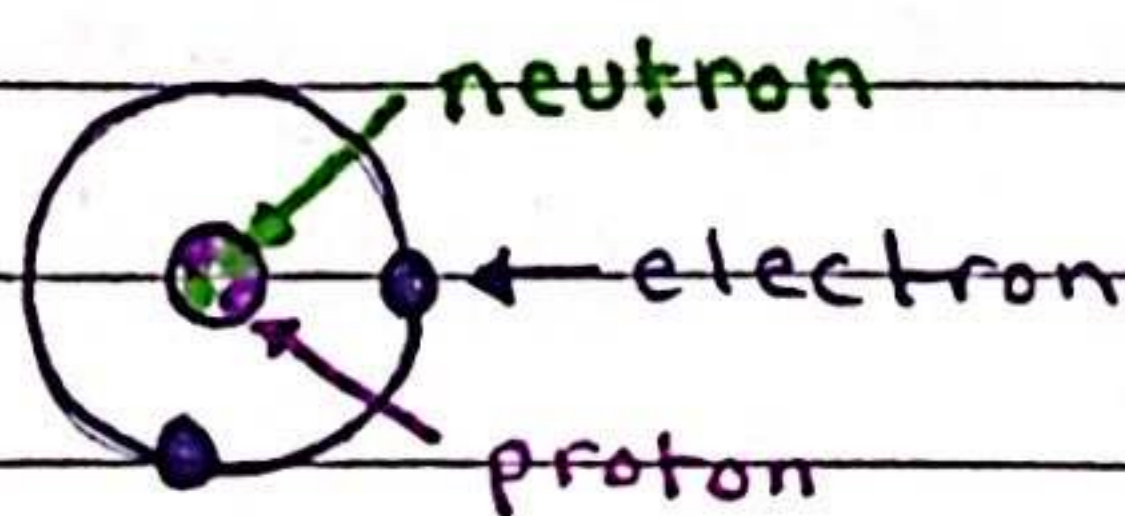
- a substance that cannot be broken down into anything simpler

Mixture

- 2 or more elements physically combined
- same properties as the elements it's made from
- can be separated by physical means

Compound

- 2 or more elements chemically combined
- different properties to the elements it's made from
- cannot be separated by physical means



relative charge

neutron :- 0

electron :- -1

proton :- +1

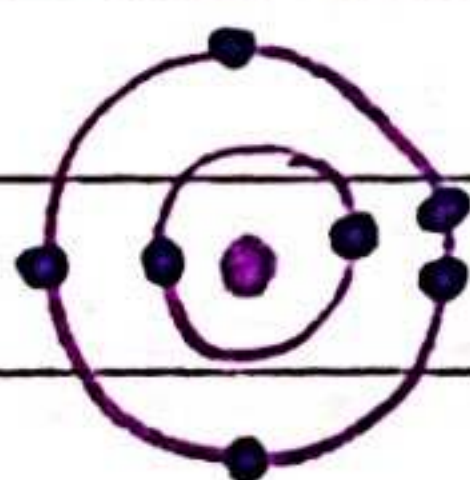
2	← atomic number
He	← atomic symbol
Helium	← name
4	← mass number

(number of protons = number of electrons)

Electronic configuration

- distribution of electrons in an atom's shell

7
N
Nitrogen
14



1st shell = 2 electrons
 2nd shell = 5 electrons

electronic configuration
 → 2, 5

- * group VIII noble gases have a full outer shell
- * number of outer shell electrons = group number
- * number of occupied electron shells = period number

Isotopes

- different atoms of the same element that have the same number of protons but a different number of neutrons
- isotopes of the same element have the same chemical properties because they have the same number of electrons

Relative atomic mass

- average weighted mass of an elements isotopes
- * total mass number of isotopes
= mass number of isotopes $\times$ abundance (%)

chlorine - 37
25%

chlorine - 35
75%

$$37 \times 25 = 925$$

$$35 \times 75 = 2625$$

$$925 + 2625 = \underline{\underline{3550}}$$

$$3550 / 100 = 35.5 \quad (\text{relative atomic mass})$$

→ ÷ by 100 because the abundance was in a percentage

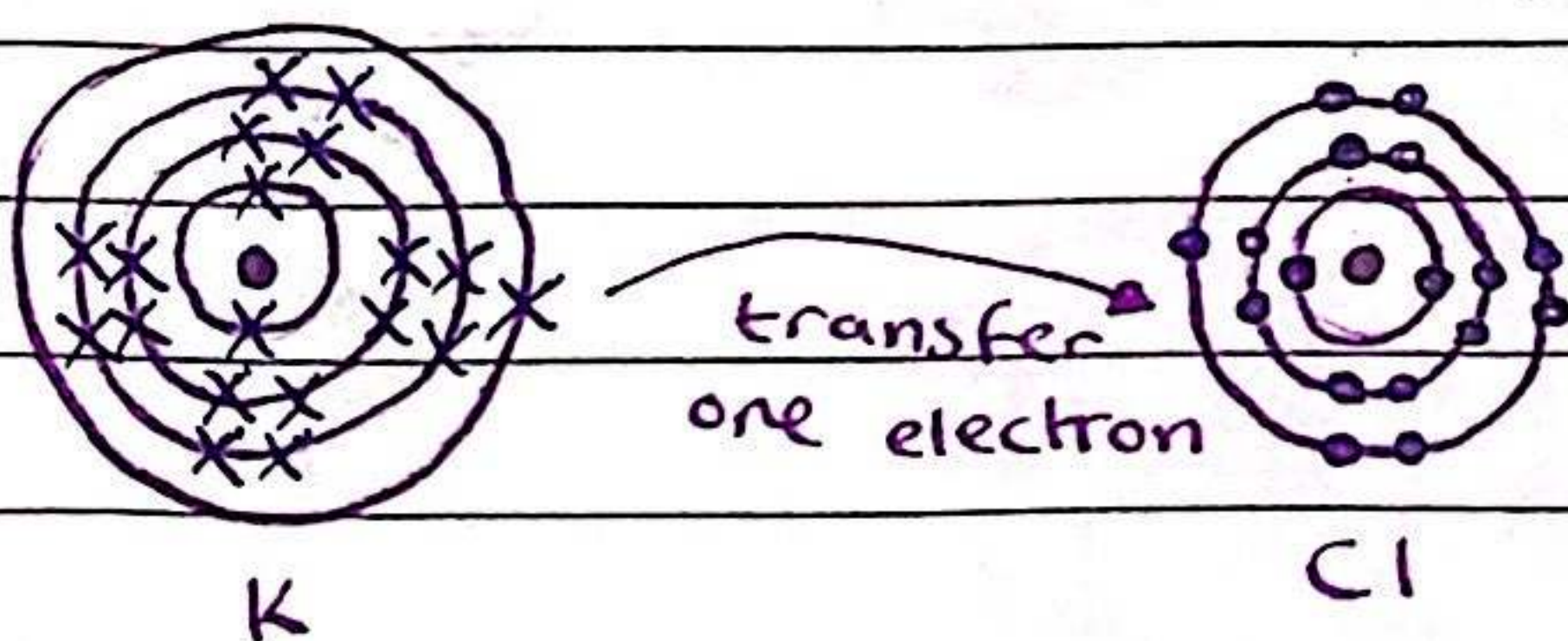
Ions

- an atom or a group of atoms in which the number of electrons is different to the number of protons
- positive ions = cations
- negative ions = anions

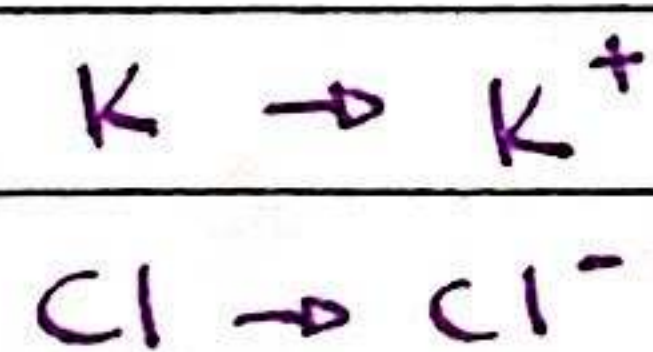
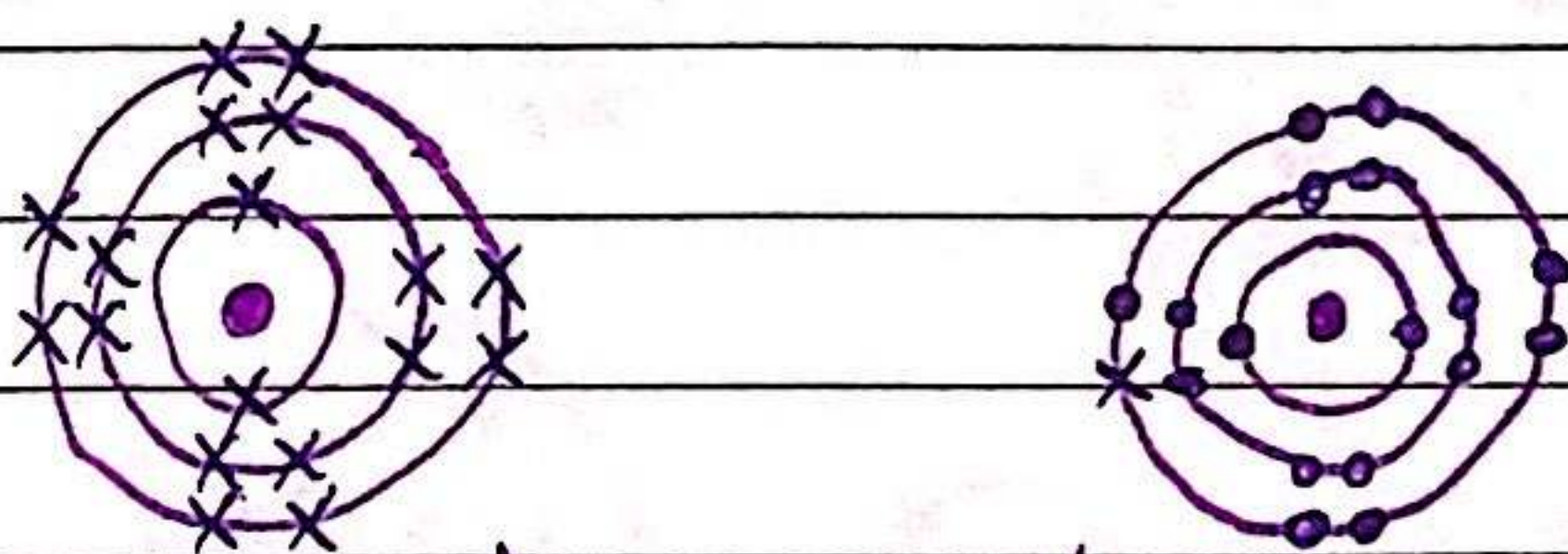
Ionic bond

- strong electrostatic attraction between oppositely charged ions
- ionic compound = a regular arrangement of alternating positive & negative ions
- giant lattice = any solid with atoms arranged in a definite pattern

* ionic bonds only happen between metals & non-metals



- all gases want to have 8 electrons in their outer shell (noble gas)

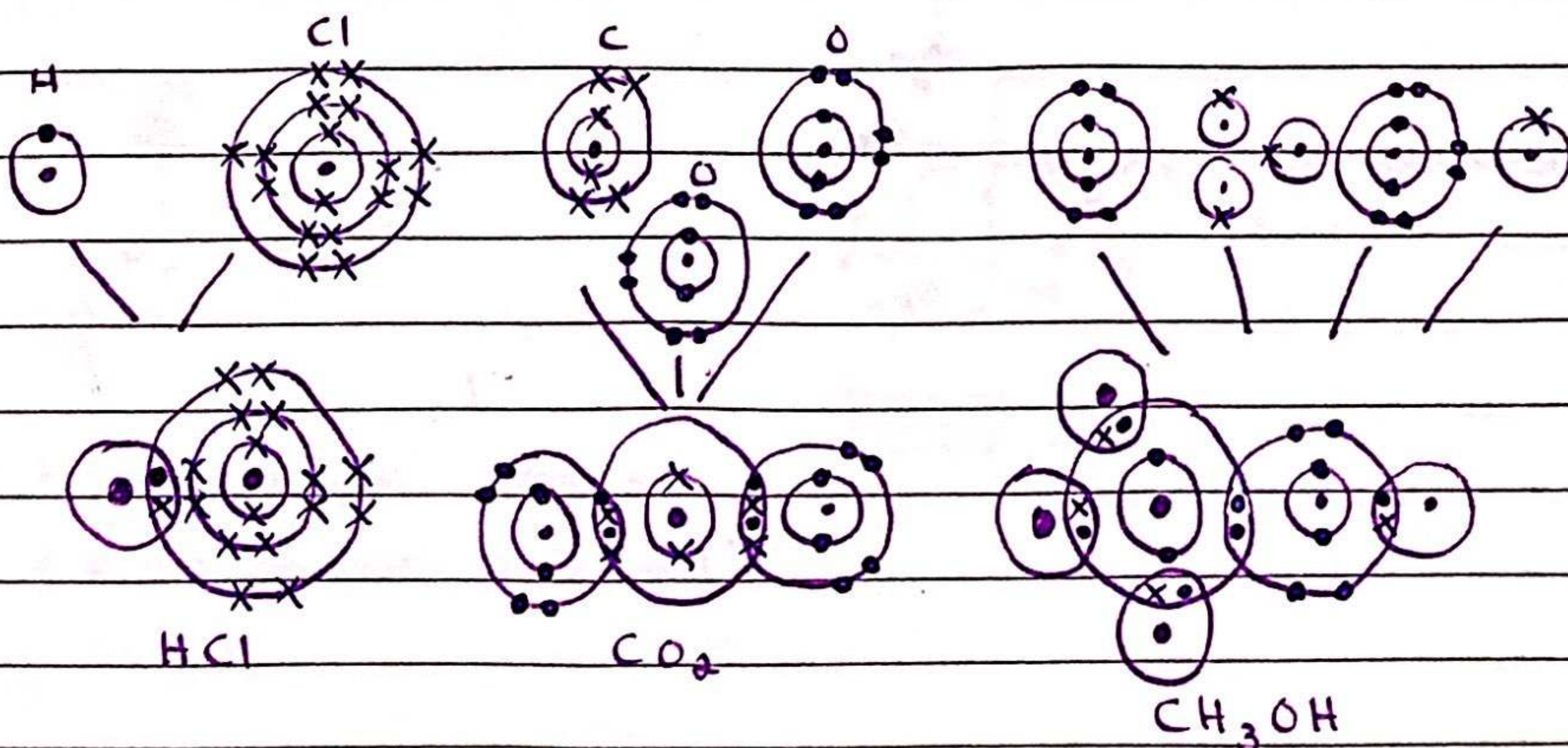
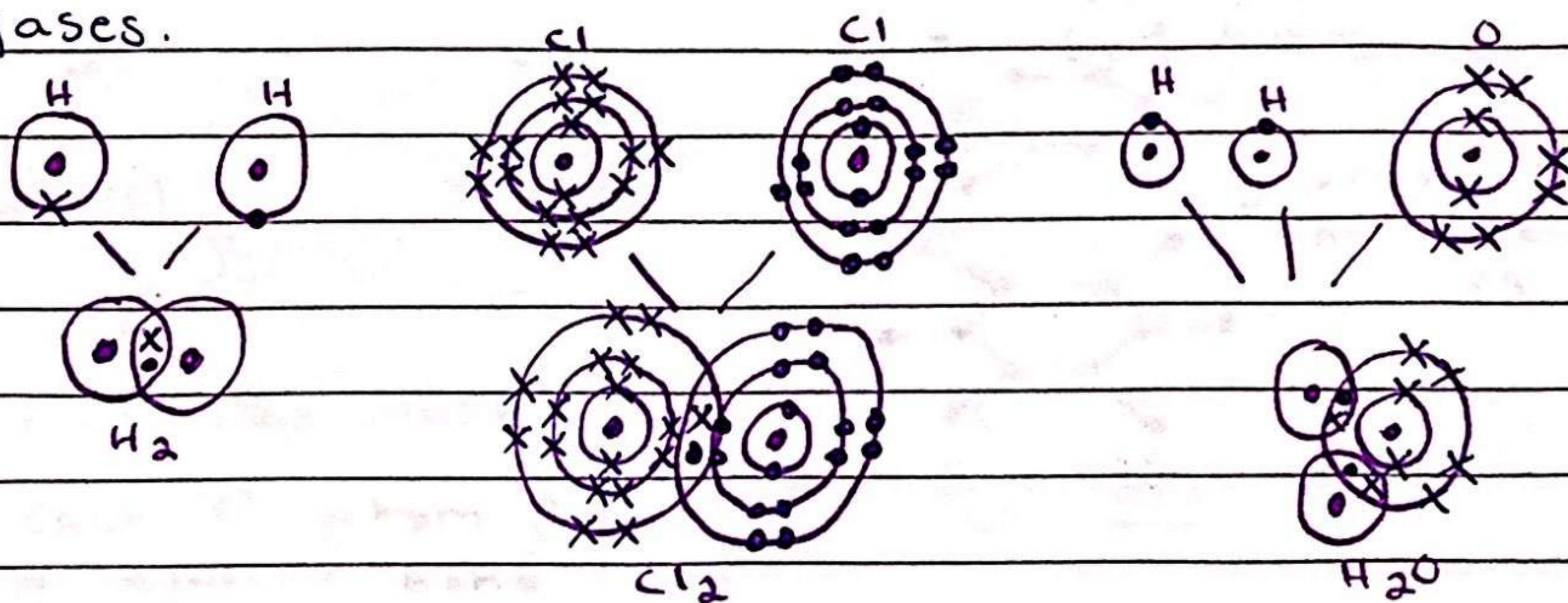


Properties of ionic compounds

- high melting / boiling points
 - the lattice contains such large number of ions, a lot of energy is needed to overcome this
- good electrical conductivity when aqueous or molten (poor when solid)
 - when aqueous or molten, the ions are free to move therefore they can conduct electricity
 - when solid, the ions are in fixed positions and cannot move, therefore they can't conduct electricity

Covalent bonds

- a bond that is formed when a pair of electrons is shared between two atoms leading to noble gases.

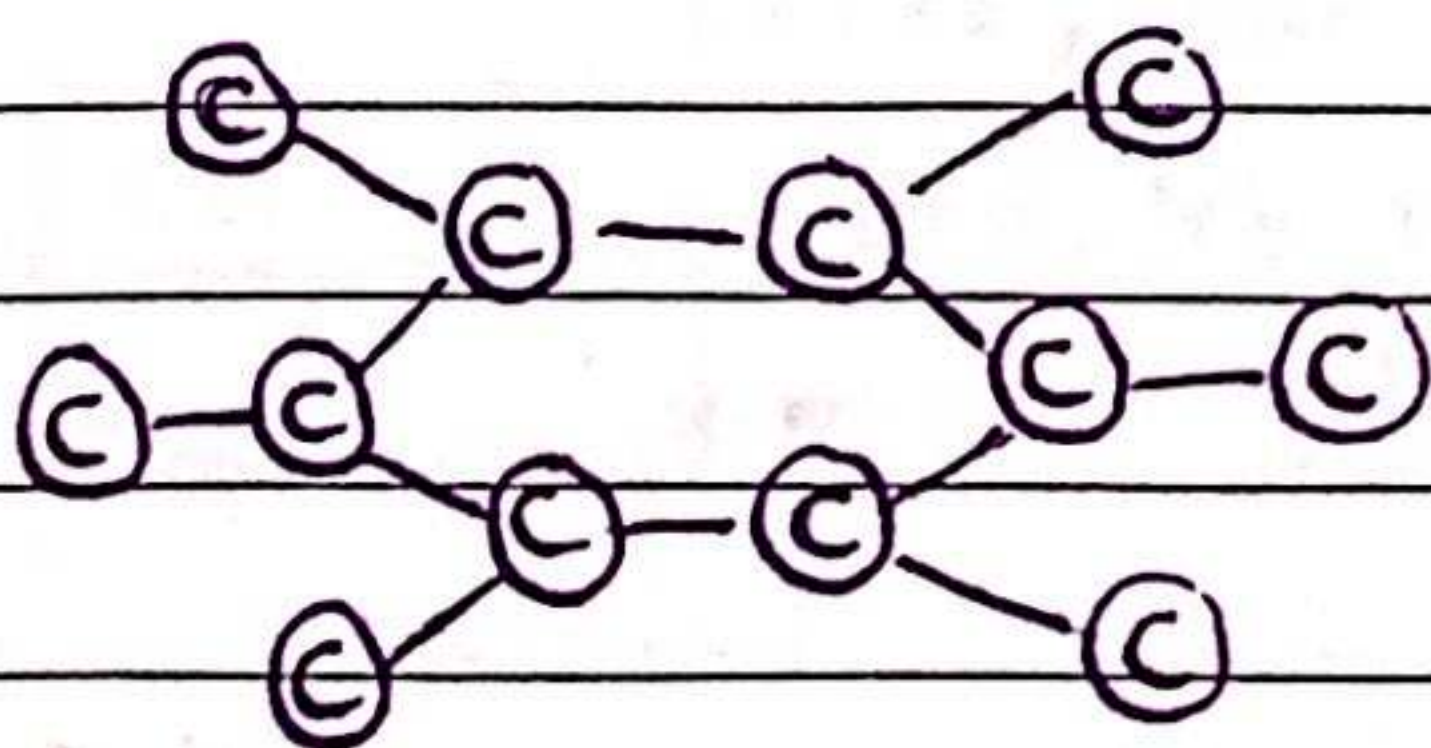


Properties of simple molecular compounds

- low melting / boiling points
 - when the substance is melted / boiled, the covalent bond is not broken, but the intermolecular force is broken
 - intermolecular force is the attractive / repulsive force between two molecules of a substance
- poor electrical conductivity
 - the molecules are uncharged, so there are no free charged particles to carry the current

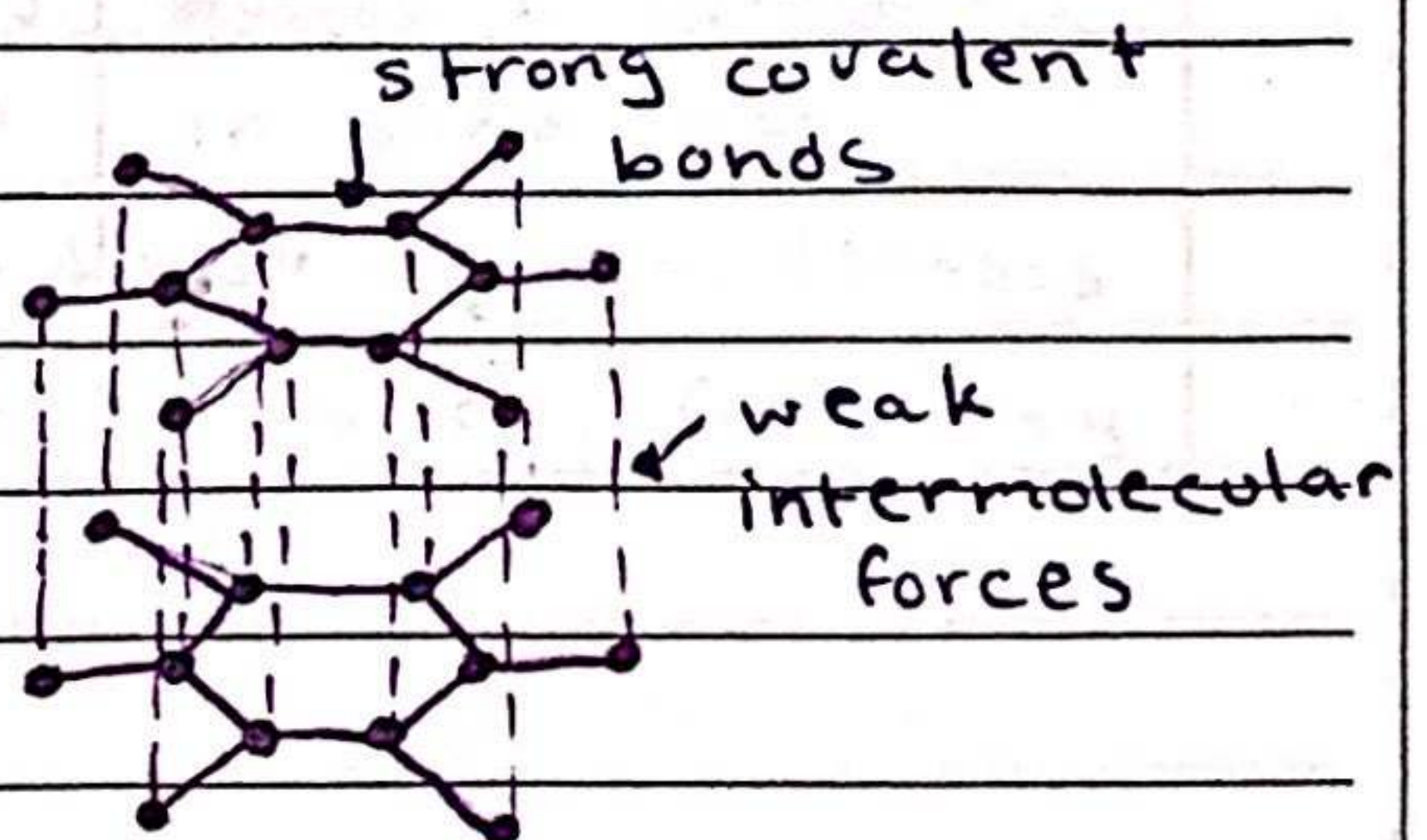
Giant covalent structures

- structure made by molecules that are joined by covalent bonds

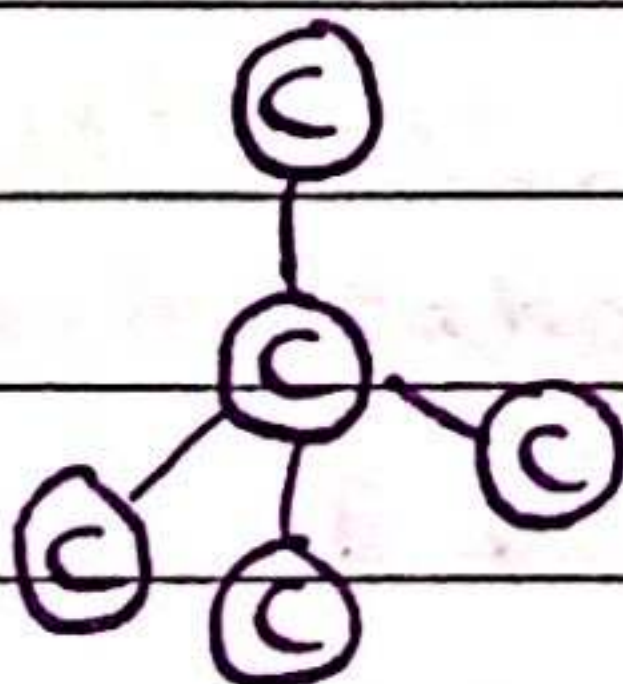


- ring-like shape

- each 'C' atom forms a covalent bond with 3 others

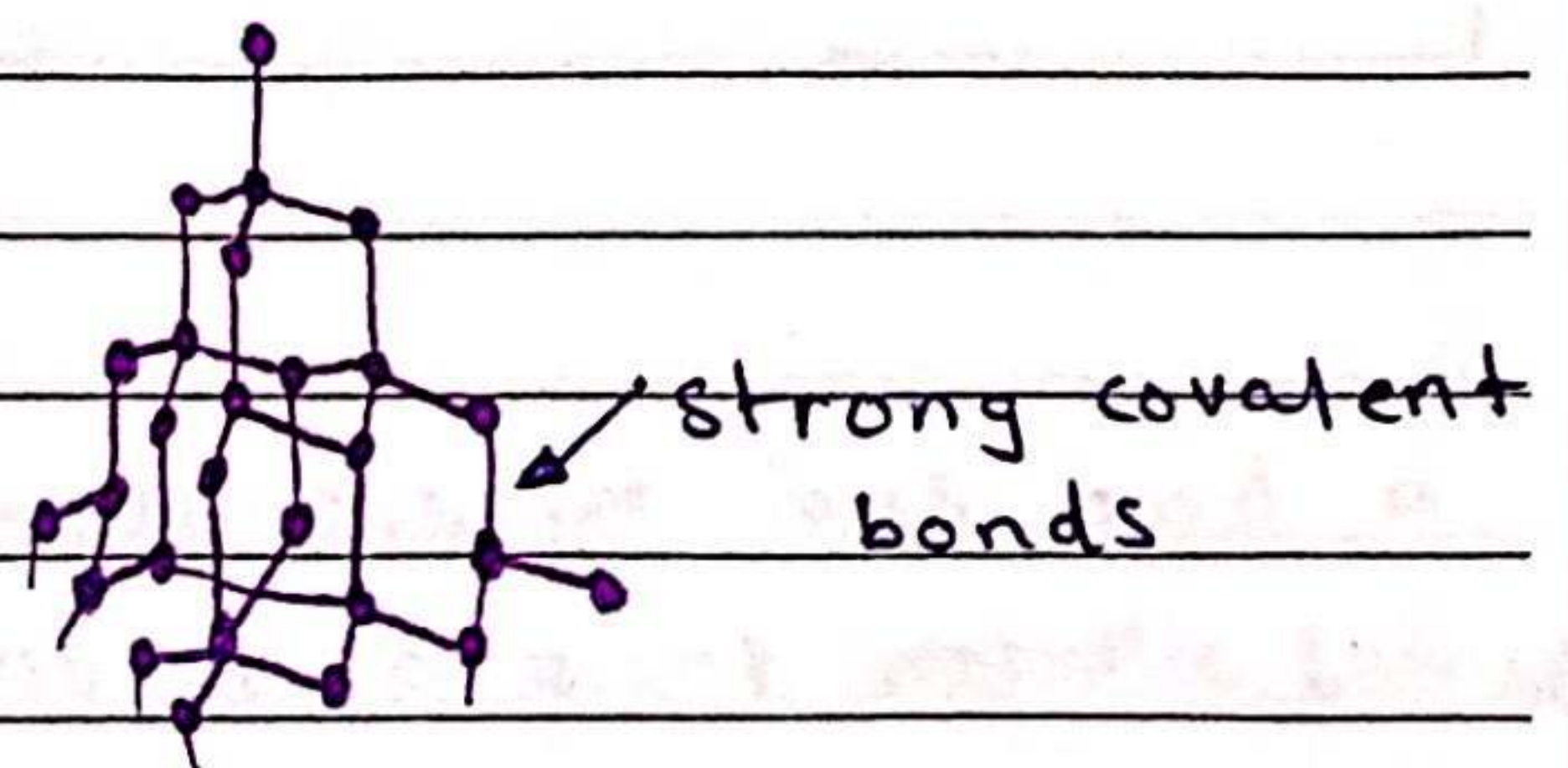


GRAPHITE

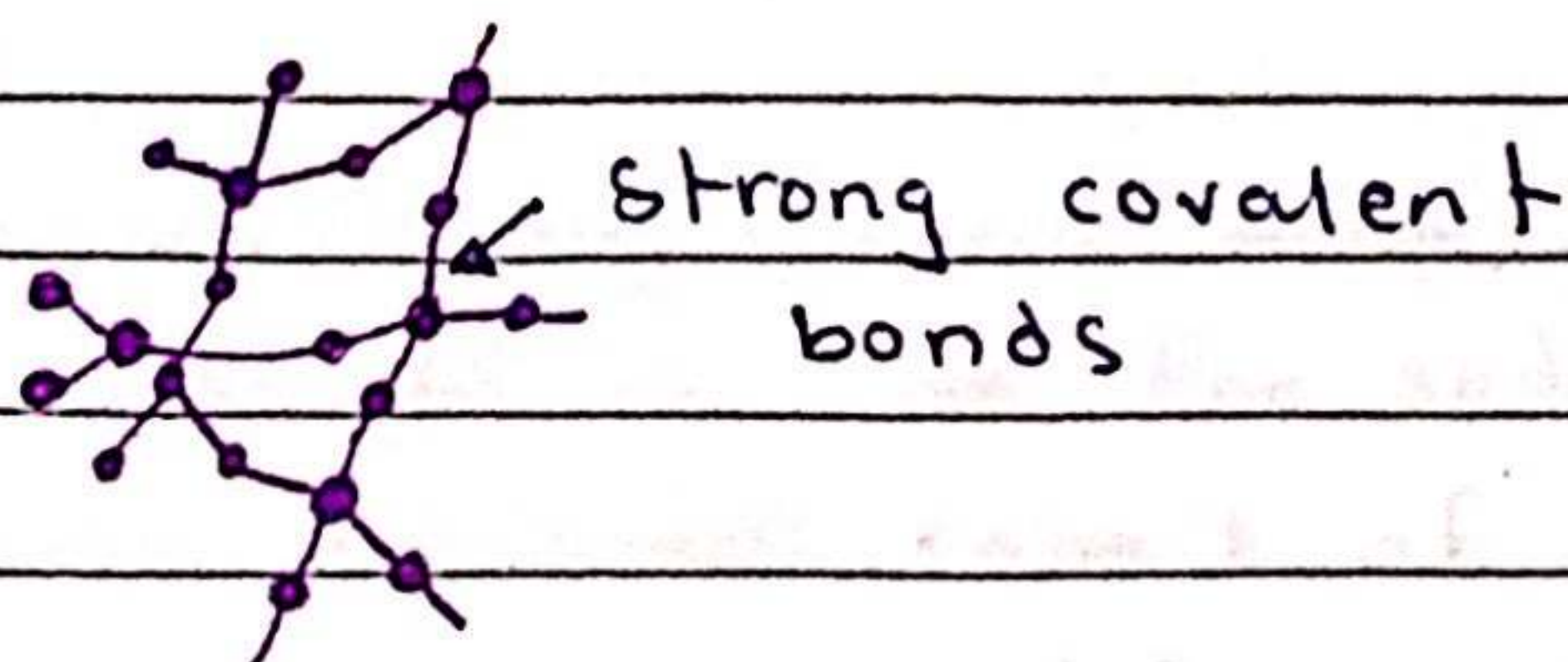
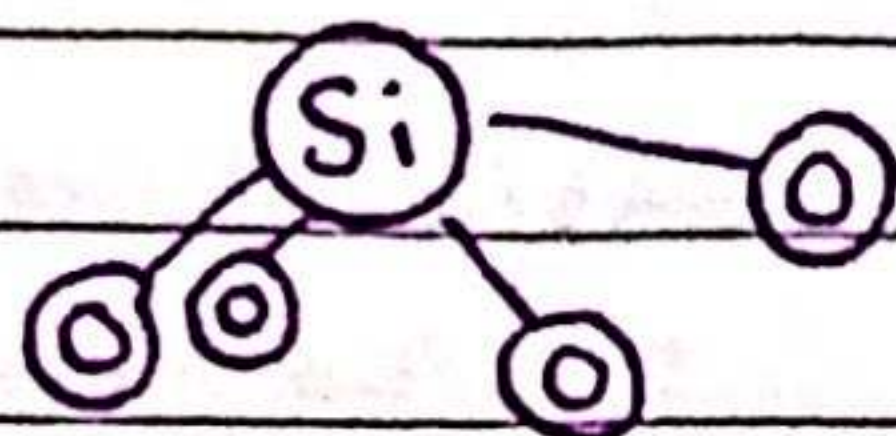


- tetrahedral shape

- melting point = 3550°C



DIAMOND



SILICA

- each silicon atom covalently bonds to 4 oxygen atoms

substance	properties	uses
<u>diamond</u>	hardest known substance doesn't conduct electricity sparkles when cut	in tools for drilling and cutting for jewellery
	soft and slippery	as lubricant for engines
<u>graphite</u>	soft and dark conducts electricity	for pencil (lead) for electrodes, connecting brushes in generators
	hard, can scratch	in sandpaper
<u>silica</u>	hard, let. light through high melting point	making glass/lenses in bricks for lining

→ diamond and silica

- ↳ high melting points due to strong covalent bonds
- ↳ form giant covalent structures

Metallic bonding

- electrostatic attraction between positive ions and a 'sea' of delocalised electrons in a giant metallic lattice

Properties of metals

- good electrical conductivity
 - delocalised electrons are free to carry electrical charge
- malleability and ductility
 - the layers of ions can slide over each other